



AP[®] Calculus BC Practice Exam

From the 2013 Administration

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A A

CALCULUS BC
SECTION I, Part A
Time—55 minutes
Number of questions—28

A CALCULATOR MAY NOT BE USED ON THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in the exam book. Do not spend too much time on any one problem.

In this exam:

- (1) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (2) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1} x = \arcsin x$).

A A

1. If $f(x) = \frac{x^2 + 3x + 2}{x + 3}$, then $f'(x) =$

(A) $2x + 3$

(B) $\frac{-x^2 - 6x - 7}{(x + 3)^2}$

(C) $\frac{x^2 + 6x + 7}{(x + 3)^2}$

(D) $\frac{x^2 + 12x + 11}{(x + 3)^2}$

(E) $\frac{3x^2 + 12x + 11}{(x + 3)^2}$

2. $\int 5x(\sqrt{x} - x^2) dx =$

(A) $\frac{15\sqrt{x}}{2} - 15x^2 + C$

(B) $5x - \frac{5x^4}{4} + C$

(C) $2x^{5/2} - \frac{5x^4}{4} + C$

(D) $\frac{25x^{5/2}}{2} - \frac{5x^4}{4} + C$

(E) $\frac{5x^{7/2}}{3} - \frac{5x^6}{6} + C$

A A

7. A population y changes at a rate modeled by the differential equation $\frac{dy}{dt} = 0.2y(1000 - y)$, where t is measured in years. What are all values of y for which the population is increasing at a decreasing rate?

(A) 500 only
(B) $0 < y < 500$ only
(C) $500 < y < 1000$ only
(D) $0 < y < 1000$
(E) $y > 1000$

-
8. Which of the following gives the length of the path described by the parametric equations $x(t) = 2 + 3t$ and $y(t) = 1 + t^2$ from $t = 0$ to $t = 1$?

(A) $\int_0^1 \sqrt{1 + \frac{4t^2}{9}} dt$
(B) $\int_0^1 \sqrt{1 + 4t^2} dt$
(C) $\int_0^1 \sqrt{3 + 3t + t^2} dt$
(D) $\int_0^1 \sqrt{9 + 4t^2} dt$
(E) $\int_0^1 \sqrt{(2 + 3t)^2 + (1 + t^2)^2} dt$

13. The graph of f' , the derivative of f , is shown in the figure above. If $f(0) = 20$, which of the following could be the value of $f(5)$?

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any part of this page is illegal.

14. If a and b are positive constants, then $\lim_{x \rightarrow \infty} \frac{\ln(bx + 1)}{\ln(ax^2 + 3)} =$

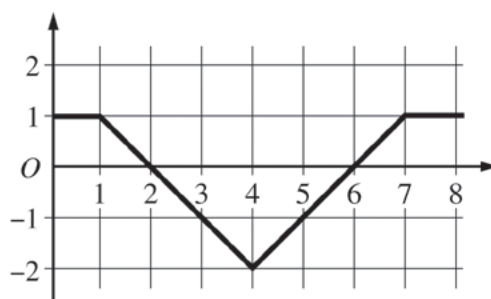
15. What are all values of x for which the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n} \left(x + \frac{3}{2}\right)^n$ converges?

- 12-

18. The graph of the function f , consisting of two line segments, is shown in the figure above. Let g be the function given by $g(x) = 2x + 1$, and let h be the function given by $h(x) = f(g(x))$. What is the value of $h'(1)$?

(A) -4 (B) -2 (C) 4 (D) 6 (E) nonexistent

A A



Graph of f

20. The graph of the function f in the figure above consists of four line segments. Let g be the function defined by

$g(x) = \int_0^x f(t) dt$. Which of the following is an equation of the line tangent to the graph of g at $x = 5$?

- (A) $y + 1 = x - 5$
- (B) $y - 2 = x - 5$
- (C) $y - 2 = -1(x - 5)$
- (D) $y + 2 = x - 5$
- (E) $y + 2 = -1(x - 5)$

A A

22. Which of the following is true about the curve $x^2 - xy + y^2 = 3$ at the point $(2, 1)$?

- (A) $\frac{dy}{dx}$ exists at $(2, 1)$, but there is no tangent line at that point.
 - (B) $\frac{dy}{dx}$ exists at $(2, 1)$, and the tangent line at that point is horizontal.
 - (C) $\frac{dy}{dx}$ exists at $(2, 1)$, and the tangent line at that point is neither horizontal nor vertical.
 - (D) $\frac{dy}{dx}$ does not exist at $(2, 1)$, and the tangent line at that point is vertical.
 - (E) $\frac{dy}{dx}$ does not exist at $(2, 1)$, and the tangent line at that point is horizontal.
-

23. What is the coefficient of x^6 in the Taylor series for $\frac{e^{3x^2}}{2}$ about $x = 0$?

- (A) $\frac{1}{1440}$
- (B) $\frac{81}{160}$
- (C) $\frac{9}{4}$
- (D) $\frac{9}{2}$
- (E) $\frac{27}{2}$

28. If $x = t^2 - 1$ and $y = \ln t$, what is $\frac{d^2y}{dx^2}$ in terms of t ?

- 21-

B**B****B****B****B****B****B****B****B****CALCULUS BC****SECTION I, Part B****Time—50 minutes****Number of questions—17**

A GRAPHING CALCULATOR IS REQUIRED FOR SOME QUESTIONS ON
THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in the exam book. Do not spend too much time on any one problem.

BE SURE YOU ARE USING PAGE 3 OF THE ANSWER SHEET TO RECORD YOUR ANSWERS TO QUESTIONS NUMBERED 76–92.

YOU MAY NOT RETURN TO PAGE 2 OF THE ANSWER SHEET.

In this exam:

- (1) The exact numerical value of the correct answer does not always appear among the choices given. When this happens, select from among the choices the number that best approximates the exact numerical value.
- (2) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (3) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1} x = \arcsin x$).

B**B****B****B****B****B****B****B****B**

76. Let f be a function whose derivative is given by $f'(x) = \ln(x^4 + 5x^3 + x^2 - 7x + 28)$. On the open interval $(-4, 1)$, at which of the following values of x does f attain a relative maximum?

(A) -3.623 only

(B) -0.871 only

(C) -3.623 and -3.284

(D) -3.459 and 0.581 only

(E) -3.459 , -0.871 , and 0.581

B**B****B****B****B****B****B****B****B**

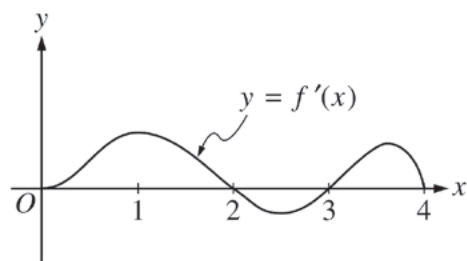
a	$\lim_{x \rightarrow a^-} f(x)$	$\lim_{x \rightarrow a^+} f(x)$	$f(a)$
-1	4	6	4
0	-3	-3	5
1	2	2	2

77. The function f has the properties indicated in the table above. Which of the following must be true?

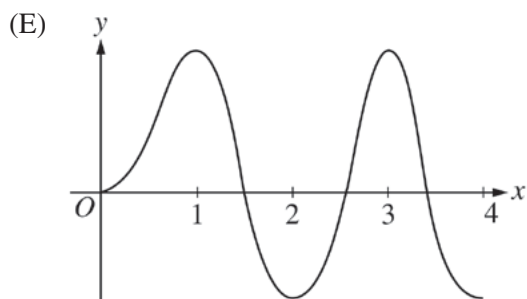
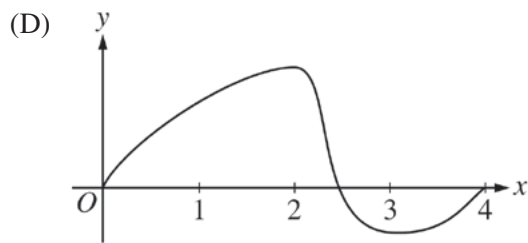
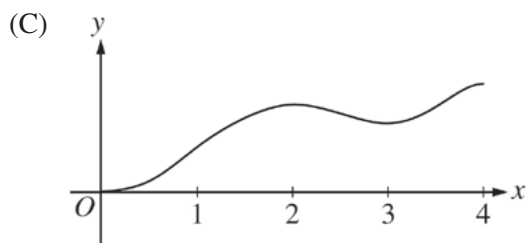
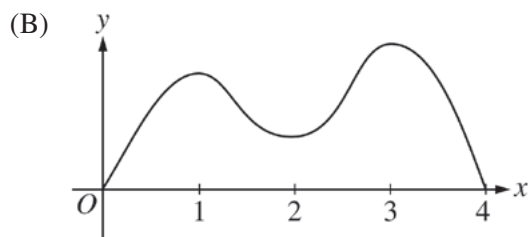
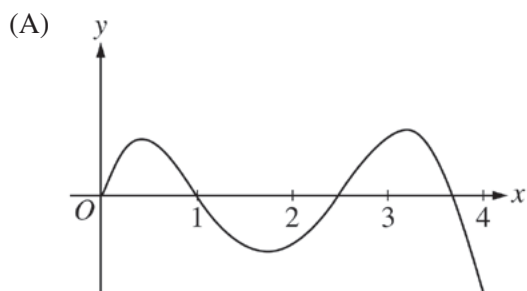
- (A) f is continuous at $x = -1$
- (B) f is continuous at $x = 0$
- (C) f is continuous at $x = 1$
- (D) f is differentiable at $x = 0$
- (E) f is differentiable at $x = 1$.

78. What is the area of the region in the first quadrant enclosed by the graphs of $y = \sin(2x)$ and $y = x$

- (A) 0.208
- (B) 0.210
- (C) 0.266
- (D) 0.660
- (E) 0.835

B**B****B****B****B****B****B****B****B**

79. The figure above shows the graph of f' , the derivative of the function f . If $f(0) = 0$, which of the following could be the graph of f ?



B**B****B****B****B****B****B****B****B**

80. The volume of a certain cone for which the sum of its radius, r , and height is constant is given by $V = \frac{1}{3}\pi r^2(10 - r)$. The rate of change of the radius of the cone with respect to time is 6. In terms of r , what is the rate of change of the volume of the cone with respect to time?

(A) $-24\pi r$ (B) $6\pi r$ (C) $\frac{20}{3}\pi r - \pi r^2$ (D) $16\pi r - \frac{4}{3}\pi r^2$ (E) $40\pi r - 6\pi r^2$

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81. A cup of tea is cooling in a room that has a constant temperature of 70 degrees Fahrenheit ($^{\circ}\text{F}$). If the initial temperature of the tea, at time $t = 0$ minutes, is 200°F and the temperature of the tea changes at the rate $R(t) = -6.89e^{-0.053t}$ degrees Fahrenheit per minute, what is the temperature, to the nearest degree, of the tea after 4 minutes?

(A) 175°F (B) 130°F (C) 95°F (D) 70°F (E) 45°F

B**B****B****B****B****B****B****B****B**

82. Consider the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$, where $a_n > 0$ and $b_n > 0$ for $n \geq 1$. If $\sum_{n=1}^{\infty} a_n$ converges, which of the following must be true?

(A) If $a_n \leq b_n$, then $\sum_{n=1}^{\infty} b_n$ converges.

(B) If $a_n \leq b_n$, then $\sum_{n=1}^{\infty} b_n$ diverges.

(C) If $b_n \leq a_n$, then $\sum_{n=1}^{\infty} b_n$ converges.

(D) If $b_n \leq a_n$, then $\sum_{n=1}^{\infty} b_n$ diverges.

(E) If $b_n \leq a_n$, then the behavior of $\sum_{n=1}^{\infty} b_n$ cannot be determined from the information given.

B**B****B****B****B****B****B****B****B**

x	0	0.5	1	1.5	2	2.5	3
$f(x)$	0	4	10	18	28	40	54

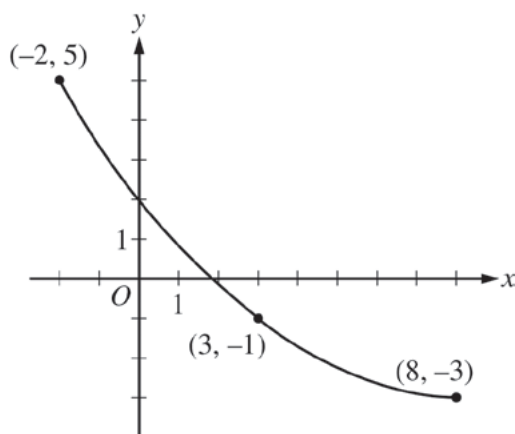
83. The table above gives selected values for a continuous function f . If f is increasing over the closed interval

$[0, 3]$, which of the following could be the value of $\int_0^3 f(x) dx$?

- (A) 50 (B) 62 (C) 77 (D) 100 (E) 154

84. Let f be a function with derivative given by $f'(x) = x^3 - 5x^2 + e^x$. On which of the following intervals is the graph of f concave down?

- (A) $(-\infty, 0.117)$ only
(B) $(-\infty, 1.144)$
(C) $(0.116, 2.062)$
(D) $(0.673, 2.863)$
(E) $(2.863, \infty)$

B**B****B****B****B****B****B****B****B**

Graph of f

85. A portion of the graph of a differentiable function f is shown above. If the value $c = 3$ satisfies the conclusion of the Mean Value Theorem applied to f on the open interval $-2 < x < 8$, what is the slope of the line tangent to the graph of f at $x = 3$?

(A) $-\frac{7}{5}$ (B) $-\frac{5}{4}$ (C) $-\frac{4}{5}$ (D) $-\frac{5}{7}$ (E) $-\frac{1}{5}$

B**B****B****B****B****B****B****B****B**

86. If $f'(x) > 0$ for all x and $f''(x) < 0$ for all x , which of the following could be a table of values for f ?

(A)

x	$f(x)$
-1	4
0	3
1	1

(B)

x	$f(x)$
-1	4
0	4
1	4

(C)

x	$f(x)$
-1	4
0	5
1	6

(D)

x	$f(x)$
-1	4
0	5
1	7

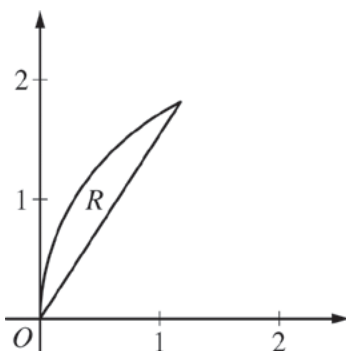
(E)

x	$f(x)$
-1	4
0	6
1	7

B**B****B****B****B****B****B****B****B**

87. The position of a particle moving in the xy -plane is given by the parametric functions $x(t)$ and $y(t)$ for which $x'(t) = t \sin t$ and $y'(t) = 5e^{-3t} + 2$. What is the slope of the line tangent to the path of the particle at the point at which $t = 2$?

- (A) 0.904 (B) 1.107 (C) 1.819 (D) 2.012 (E) 3.660

B**B****B****B****B****B****B****B****B**

88. Let R be the region in the first quadrant that is bounded above by the polar curve $r = 4\cos\theta$ and below by the line $\theta = 1$, as shown in the figure above. What is the area of R ?
- (A) 0.317 (B) 0.465 (C) 0.929 (D) 2.618 (E) 5.819

-
89. What is the volume of the solid generated when the region bounded by the graph of $x = \sqrt{y-2}$ and the lines $x = 0$ and $y = 5$ is revolved about the y -axis?
- (A) 3.464 (B) 4.500 (C) 7.854 (D) 10.883 (E) 14.137

B**B****B****B****B****B****B****B****B**

90. Which of the following statements are true about the series $\sum_{n=2}^{\infty} a_n$, where $a_n = \frac{(-1)^n}{\sqrt{n} + (-1)^n}$?

I. The series is alternating.

II. $|a_{n+1}| \leq |a_n|$ for all $n \geq 2$

III. $\lim_{n \rightarrow \infty} a_n = 0$

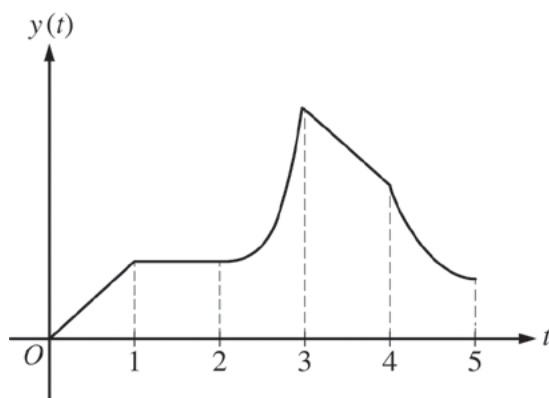
(A) None

(B) I only

(C) I and II only

(D) I and III only

(E) I, II, and III

B**B****B****B****B****B****B****B****B**

91. A particle moves along the y -axis. The graph of the particle's position $y(t)$ at time t is shown above for $0 \leq t \leq 5$. For what values of t is the velocity of the particle negative and the acceleration positive?

- (A) $0 < t < 1$ (B) $1 < t < 2$ (C) $2 < t < 3$ (D) $3 < t < 4$ (E) $4 < t < 5$

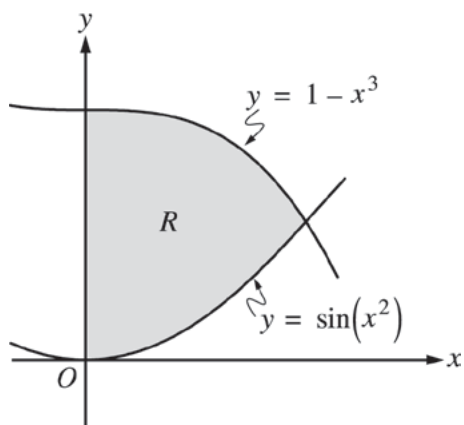
92. If f is a function such that $f'(x) = -f(x)$, then $\int x f(x) dx =$

- (A) $f(x)(x+1) + C$
(B) $-f(x)(x+1) + C$
(C) $\frac{x^2}{2} f(x) + C$
(D) $-\frac{x^2}{2} f(x) + C$
(E) $-\frac{x^2}{2} f(x) \left(1 + \frac{x}{3}\right) + C$

CALCULUS BC
SECTION II, Part A
Time—30 minutes
Number of problems—2

A graphing calculator is required for these problems.

GO ON TO THE NEXT PAGE.



1. Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$, as shown in the figure above.
- (a) Find the area of R .

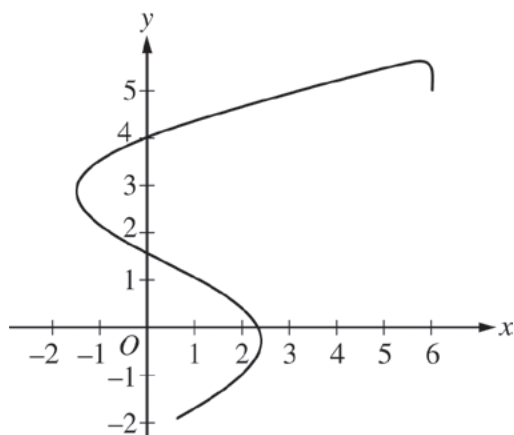
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1**1****1****1****1****1****1****1****1****1**

- (b) A horizontal line, $y = k$, is drawn through the point where the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect. Find k and determine whether this line divides R into two regions of equal area. Show the work that leads to your conclusion.

-
- (c) Find the volume of the solid generated when R is revolved about the line $y = -3$.



2. A planetary rover travels on a flat surface. The path of the rover for the time interval $0 \leq t \leq 2$ hours is shown in the rectangular coordinate system above. The rover starts at the point with coordinates $(6, 5)$ at time $t = 0$. The coordinates $(x(t), y(t))$ of the position of the rover change at rates given by

$$\begin{aligned} x'(t) &= -12 \sin(2t^2) \\ y'(t) &= 10 \cos(1 + \sqrt{t}), \end{aligned}$$

where $x(t)$ and $y(t)$ are measured in meters and t is measured in hours.

- (a) Find the acceleration vector of the rover at time $t = 1$. Find the speed of the rover at time $t = 1$.

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(b) Find the total distance that the rover travels over the time interval $0 \leq t \leq 1$.

(c) Find the y-coordinate of the position of the rover at time $t = 1$.

(d) The rover receives a signal at each point where the line tangent to its path has slope $\frac{1}{2}$. At what times t , for $0 \leq t \leq 2$, does the rover receive a signal?

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CALCULUS BC
SECTION II, Part B
Time—60 minutes
Number of problems—4

No calculator is allowed for these problems.

DO NOT BREAK THE SEALS UNTIL YOU ARE TOLD TO DO SO.

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NO CALCULATOR ALLOWED

t (days)	0	10	22	30
$W'(t)$ (GL per day)	0.6	0.7	1.0	0.5

3. The twice-differentiable function W models the volume of water in a reservoir at time t , where $W(t)$ is measured in gigaleters (GL) and t is measured in days. The table above gives values of $W'(t)$ sampled at various times during the time interval $0 \leq t \leq 30$ days. At time $t = 30$, the reservoir contains 125 gigaleters of water.
- (a) Use the tangent line approximation to W at time $t = 30$ to predict the volume of water $W(t)$, in gigaleters, in the reservoir at time $t = 32$. Show the computations that lead to your answer.

- (b) Use a left Riemann sum, with the three subintervals indicated by the data in the table, to approximate $\int_0^{30} W'(t) dt$. Use this approximation to estimate the volume of water $W(t)$, in gigaleters, in the reservoir at time $t = 0$. Show the computations that lead to your answer.

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NO CALCULATOR ALLOWED

- (c) Explain why there must be at least one time t , other than $t = 10$, such that $W'(t) = 0.7$ GL/day.

-
- (d) The equation $A = 0.3W^{2/3}$ gives the relationship between the area A , in square kilometers, of the surface of the reservoir, and the volume of water $W(t)$, in giga-liters, in the reservoir. Find the instantaneous rate of change of A , in square kilometers per day, with respect to t when $t = 30$ days.

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NO CALCULATOR ALLOWED

4. Consider the function f given by $f(x) = xe^{-x^2}$ for all real numbers x .

(a) At what value of x does $f(x)$ attain its absolute maximum? Justify your answer.

(b) Find an antiderivative of f .

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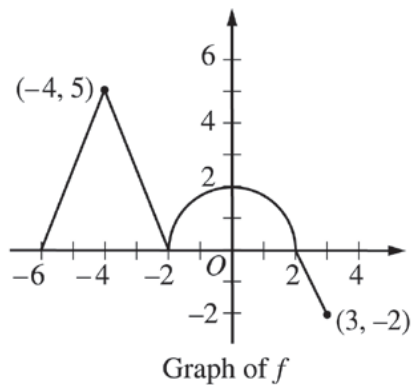
NO CALCULATOR ALLOWED

(c) Find the value of $\int_0^\infty xf(x)dx$, given the fact that $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$.

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NO CALCULATOR ALLOWED



5. The graph of the continuous function f , consisting of three line segments and a semicircle, is shown above.

Let g be the function given by $g(x) = \int_{-2}^x f(t) dt$.

(a) Find $g(-6)$ and $g(3)$.

(b) Find $g'(0)$.

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NO CALCULATOR ALLOWED

- (c) Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a horizontal tangent. Determine whether g has a local maximum, a local minimum, or neither at each of these values. Justify your answers.

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- (d) Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a point of inflection. Explain your reasoning.

NO CALCULATOR ALLOWED

6. The function f satisfies the equation

$$f'(x) = f(x) + x + 1$$

and $f(0) = 2$. The Taylor series for f about $x = 0$ converges to $f(x)$ for all x .

- (a) Write an equation for the line tangent to the curve $y = f(x)$ at the point where $x = 0$.

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- (b) Find $f''(0)$ and find the second-degree Taylor polynomial for f about $x = 0$.

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NO CALCULATOR ALLOWED

(c) Find the fourth-degree Taylor polynomial for f about $x = 0$.

(d) Find $f^{(n)}(0)$, the n th derivative of f at $x = 0$, for $n \geq 2$. Use the Taylor series for f about $x = 0$ and the Taylor series for e^x about $x = 0$ to find a polynomial expression for $f(x) - 4e^x$.

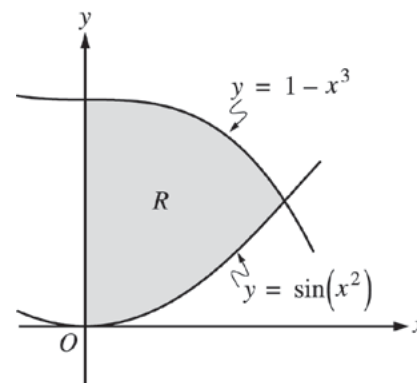
Answer Key for AP Calculus BC Practice Exam, Section I

Question 1: C	Question 24: A
Question 2: C	Question 25: C
Question 3: D	Question 26: D
Question 4: E	Question 27: A
Question 5: A	Question 28: A
Question 6: D	Question 76: A
Question 7: C	Question 77: C
Question 8: D	Question 78: B
Question 9: D	Question 79: C
Question 10: D	Question 80: E
Question 11: D	Question 81: A
Question 12: E	Question 82: C
Question 13: D	Question 83: B
Question 14: B	Question 84: C
Question 15: B	Question 85: C
Question 16: A	Question 86: E
Question 17: B	Question 87: B
Question 18: A	Question 88: B
Question 19: C	Question 89: E
Question 20: E	Question 90: D
Question 21: A	Question 91: E
Question 22: D	Question 92: B
Question 23: C	

AP[®] CALCULUS BC
2013 SCORING GUIDELINES

Question 1

Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$, as shown in the figure above.



- (a) Find the area of R .
- (b) A horizontal line, $y = k$, is drawn through the point where the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect. Find k and determine whether this line divides R into two regions of equal area. Show the work that leads to your conclusion.
- (c) Find the volume of the solid generated when R is revolved about the line $y = -3$.

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$.

(a)
$$\text{Area} = \int_0^A (1 - x^3 - \sin(x^2)) dx$$

$$= 0.533 \text{ (or } 0.534)$$

(b) $k = B = 0.552352$

$$\int_0^A (1 - x^3 - k) dx = 0.257 \text{ (or } 0.256)$$

$$\int_0^A (k - \sin(x^2)) dx = 0.277 \text{ (or } 0.276)$$

The two regions have unequal areas.

(c)
$$\text{Volume} = \pi \int_0^A \left((1 - x^3 + 3)^2 - (\sin(x^2) + 3)^2 \right) dx$$

$$= 11.841 \text{ (or } 11.840)$$

1 : correct limits in an integral in (a), (b), or (c)

2 : $\begin{cases} 1 : \text{integrand} \\ 1 : \text{answer} \end{cases}$

3 : $\begin{cases} 1 : \text{integral(s) with } k \text{ value} \\ 1 : \text{value(s) of integral(s)} \\ 1 : \text{conclusion tied to part (a)} \\ \text{or comparison of two integrals} \end{cases}$

Note: Stating k value only does not earn a point.

3 : $\begin{cases} 2 : \text{integrand} \\ 1 : \text{answer} \end{cases}$

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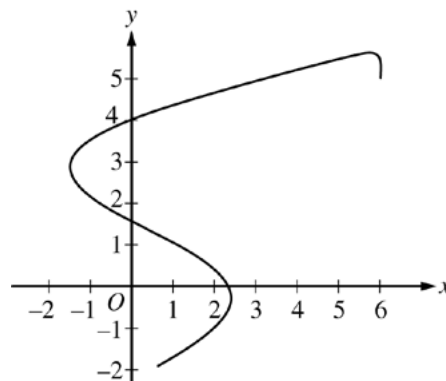
Question 2

A planetary rover travels on a flat surface. The path of the rover for the time interval $0 \leq t \leq 2$ hours is shown in the rectangular coordinate system above. The rover starts at the point with coordinates $(6, 5)$ at time $t = 0$. The coordinates $(x(t), y(t))$ of the position of the rover change at rates given by

$$x'(t) = -12 \sin(2t^2)$$

$$y'(t) = 10 \cos(1 + \sqrt{t}),$$

where $x(t)$ and $y(t)$ are measured in meters and t is measured in hours.



- (a) Find the acceleration vector of the rover at time $t = 1$. Find the speed of the rover at time $t = 1$.
- (b) Find the total distance that the rover travels over the time interval $0 \leq t \leq 1$.
- (c) Find the y -coordinate of the position of the rover at time $t = 1$.
- (d) The rover receives a signal at each point where the line tangent to its path has slope $\frac{1}{2}$. At what times t , for $0 \leq t \leq 2$, does the rover receive a signal?

(a) $a(1) = \langle x''(1), y''(1) \rangle \quad \langle 19.975, -4.546 \rangle$

$$\text{Speed} = \sqrt{(x'(1))^2 + (y'(1))^2} = 11.678$$

2 : $\begin{cases} 1 : \text{acceleration vector} \\ 1 : \text{speed} \end{cases}$

(b) $\text{Distance} = \int_0^1 \sqrt{(x'(t))^2 + (y'(t))^2} dt = 6.704 \text{ (or } 6.703)$

3 : $\begin{cases} 2 : \text{integral} \\ 1 : \text{answer} \end{cases}$

(c) $y(1) = 5 + \int_0^1 y'(t) dt = 4.057 \text{ (or } 4.056)$

2 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$

(d) $\frac{dy}{dx} = \frac{y'(t)}{x'(t)} = \frac{10 \cos(1 + \sqrt{t})}{-12 \sin(2t^2)} = \frac{1}{2}$

2 : $\begin{cases} 1 : \text{equation} \\ 1 : \text{answer} \end{cases}$

$$t = 1.072$$

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Question 3

t (days)	0	10	22	30
$W'(t)$ (GL per day)	0.6	0.7	1.0	0.5

The twice-differentiable function W models the volume of water in a reservoir at time t , where $W(t)$ is measured in giga­liters (GL) and t is measured in days. The table above gives values of $W'(t)$ sampled at various times during the time interval $0 \leq t \leq 30$ days. At time $t = 30$, the reservoir contains 125 giga­liters of water.

- (a) Use the tangent line approximation to W at time $t = 30$ to predict the volume of water $W(t)$, in giga­liters, in the reservoir at time $t = 32$. Show the computations that lead to your answer.
- (b) Use a left Riemann sum, with the three subintervals indicated by the data in the table, to approximate $\int_0^{30} W'(t) dt$. Use this approximation to estimate the volume of water $W(t)$, in giga­liters, in the reservoir at time $t = 0$. Show the computations that lead to your answer.
- (c) Explain why there must be at least one time t , other than $t = 10$, such that $W'(t) = 0.7$ GL/day.
- (d) The equation $A = 0.3W^{2/3}$ gives the relationship between the area A , in square kilometers, of the surface of the reservoir, and the volume of water $W(t)$, in giga­liters, in the reservoir. Find the instantaneous rate of change of A , in square kilometers per day, with respect to t when $t = 30$ days.

- (a) An equation of the tangent line is $y = 0.5(t - 30) + 125$.
 $W(32) \approx 0.5(32 - 30) + 125 = 126$

1 : answer

- (b) $\int_0^{30} W'(t) dt \approx (10)(0.6) + (12)(0.7) + (8)(1.0) = 22.4$
 $W(0) = W(30) - \int_0^{30} W'(t) dt = 125 - 22.4 = 102.6$

3 : $\begin{cases} 1 : \text{left Riemann sum} \\ 1 : \text{approximation} \\ 1 : \text{answer} \end{cases}$

- (c) W' is differentiable $\Rightarrow W'$ is continuous.

$$W'(30) = 0.5 < 0.7 < 1.0 = W'(22)$$

By the Intermediate Value Theorem, there must be at least one time t , $22 \leq t \leq 30$, such that $W'(t) = 0.7$.

2 : explanation

- (d) $\frac{dA}{dt} = (0.3)\frac{2}{3}W^{-1/3} \cdot \frac{dW}{dt} = \frac{0.2}{\sqrt[3]{W}} \cdot \frac{dW}{dt}$

$$\left. \frac{dA}{dt} \right|_{t=30} = \frac{0.2}{\sqrt[3]{125}} \cdot 0.5 = 0.02$$

3 : $\begin{cases} 2 : \frac{dA}{dt} \\ 1 : \text{answer} \end{cases}$

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Question 4

$$f(x) = xe^{-x^2} \text{ for all real numbers } x.$$

- (a) At what value of x does $f(x)$ attain its absolute maximum? Justify your answer.
- (b) Find an antiderivative of f .
- (c) Find the value of $\int_0^\infty xf(x) dx$, given the fact that $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$.

(a) $f'(x) = e^{-x^2} - 2x^2e^{-x^2} = (1 - 2x^2)e^{-x^2}$

$$f'(x) = 0 \text{ when } x = -\frac{1}{\sqrt{2}} \text{ or } x = \frac{1}{\sqrt{2}}$$

$f(x) < 0$ for $x < 0$, and $f(x) > 0$ for $x > 0$. Therefore, a maximum can only occur at a positive x -value.

$f'(x)$ changes sign from positive to negative at $x = \frac{1}{\sqrt{2}}$, so

f has a relative maximum at $x = \frac{1}{\sqrt{2}}$.

f increases on $0 < x \leq \frac{1}{\sqrt{2}}$ and decreases on $\frac{1}{\sqrt{2}} \leq x < \infty$.

Therefore, there is an absolute maximum at $x = \frac{1}{\sqrt{2}}$.

(b) $\int xe^{-x^2} dx = -\frac{1}{2}e^{-x^2} + C$

(c) Using integration by parts,

$$u = x \quad dv = xe^{-x^2} dx$$

$$du = dx \quad v = -\frac{1}{2}e^{-x^2}$$

$$\int xf(x) dx = \int x^2e^{-x^2} dx = -\frac{1}{2}xe^{-x^2} + \frac{1}{2}\int e^{-x^2} dx$$

Therefore,

$$\begin{aligned} \int_0^\infty xf(x) dx &= \lim_{b \rightarrow \infty} \left[-\frac{1}{2}xe^{-x^2} \Big|_0^b + \frac{1}{2}\int_0^b e^{-x^2} dx \right] \\ &= \lim_{b \rightarrow \infty} \left(-\frac{1}{2}be^{-b^2} \right) + \frac{1}{2}\int_0^\infty e^{-x^2} dx \\ &= 0 + \frac{1}{2} \cdot \frac{\sqrt{\pi}}{2} = \frac{\sqrt{\pi}}{4} \end{aligned}$$

$$5 : \begin{cases} 2 : f'(x) \\ 1 : \text{solves } f'(x) = 0 \\ 1 : \text{answer} \\ 1 : \text{justification} \end{cases}$$

1 : antiderivative

$$3 : \begin{cases} 2 : \text{integration by parts} \\ 1 : \text{answer} \end{cases}$$

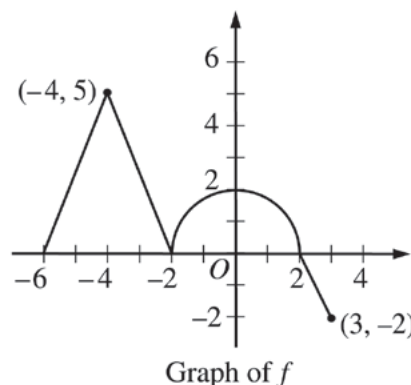
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Question 5

The graph of the continuous function f , consisting of three line segments and a semicircle, is shown above. Let g be the function given by

$$g(x) = \int_{-2}^x f(t) \, dt.$$

- (a) Find $g(-6)$ and $g(3)$.
- (b) Find $g'(0)$.
- (c) Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a horizontal tangent. Determine whether g has a local maximum, a local minimum, or neither at each of these values. Justify your answers.
- (d) Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a point of inflection. Explain your reasoning.



(a) $g(-6) = \int_{-2}^{-6} f(t) \, dt = -\int_{-6}^{-2} f(t) \, dt = -\frac{1}{2} \cdot 4 \cdot 5 = -10$

$$g(3) = \int_{-2}^3 f(t) \, dt = \frac{1}{2} \pi \cdot 2^2 - \frac{1}{2} \cdot 1 \cdot 2 = 2\pi - 1$$

(b) $g'(0) = f(0) = 2$

- (c) The graph of g has a horizontal tangent at $x = -2$ and $x = 2$ where $g'(x) = f(x) = 0$.

The graph of g has neither a local maximum nor a local minimum at $x = -2$ because $g'(x) = f(x)$ does not change sign at $x = -2$.

The graph of g has a local maximum at $x = 2$ because $g'(x) = f(x)$ changes sign from positive to negative at $x = 2$.

- (d) The graph of g has a point of inflection at $x = -4$, $x = -2$, and $x = 0$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -4$ and $x = 0$, and changes from decreasing to increasing at $x = -2$.

OR

$g''(x) = f'(x)$ changes from positive to negative at $x = -4$ and $x = 0$, and changes from negative to positive at $x = -2$.

$$2 : \begin{cases} 1 : g(-6) \\ 1 : g(3) \end{cases}$$

$$1 : g'(0)$$

$$3 : \begin{cases} 1 : \text{horizontal tangent at } x = -2 \\ \text{and } x = 2 \\ 2 : \text{answers with justifications} \end{cases}$$

$$3 : \begin{cases} 2 : \text{values of } x \\ 1 : \text{explanation} \end{cases}$$

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Question 6

The function f satisfies the equation

$$f'(x) = f(x) + x + 1$$

and $f(0) = 2$. The Taylor series for f about $x = 0$ converges to $f(x)$ for all x .

- (a) Write an equation for the line tangent to the curve $y = f(x)$ at the point where $x = 0$.
- (b) Find $f''(0)$ and find the second-degree Taylor polynomial for f about $x = 0$.
- (c) Find the fourth-degree Taylor polynomial for f about $x = 0$.
- (d) Find $f^{(n)}(0)$, the n th derivative of f at $x = 0$, for $n \geq 2$. Use the Taylor series for f about $x = 0$ and the Taylor series for e^x about $x = 0$ to find a polynomial expression for $f(x) - 4e^x$.

- (a) $f'(0) = f(0) + 0 + 1 = 3$
 An equation for the tangent line is $y = 3x + 2$.

1 : tangent line equation

- (b) $f''(x) = f'(x) + 1$; $f''(0) = f'(0) + 1 = 3 + 1 = 4$

2 : $\begin{cases} 1 : f''(0) \\ 1 : \text{second-degree Taylor polynomial} \end{cases}$

$$P_2(x) = 2 + 3x + \frac{4}{2!}x^2 = 2 + 3x + 2x^2$$

- (c) $f'''(x) = f''(x)$; $f'''(0) = f''(0) = 4$

2 : $\begin{cases} 1 : f'''(0) \text{ and } f^{(4)}(0) \\ 1 : \text{fourth-degree Taylor polynomial} \end{cases}$

$$f^{(4)}(x) = f'''(x); f^{(4)}(0) = f'''(0) = 4$$

$$\begin{aligned} P_4(x) &= 2 + 3x + \frac{4}{2!}x^2 + \frac{4}{3!}x^3 + \frac{4}{4!}x^4 \\ &= 2 + 3x + 2x^2 + \frac{2}{3}x^3 + \frac{1}{6}x^4 \end{aligned}$$

- (d) $f^{(n)}(0) = 4$ for $n \geq 2$

$$f(x) = 2 + 3x + \frac{4}{2!}x^2 + \frac{4}{3!}x^3 + \frac{4}{4!}x^4 + \frac{4}{5!}x^5 + \dots$$

$$4e^x = 4 + 4x + \frac{4}{2!}x^2 + \frac{4}{3!}x^3 + \frac{4}{4!}x^4 + \frac{4}{5!}x^5 + \dots$$

$$\text{Therefore, } f(x) - 4e^x = -2 - x.$$

4 : $\begin{cases} 1 : f^{(n)}(0) \text{ for } n \geq 2 \\ 1 : \text{Taylor series for } f \\ 1 : \text{Taylor series for } e^x \\ 1 : \text{polynomial expression} \end{cases}$